

Ultimatum Game Student Handout

TOK Cognitive Science Activity Bank • Human Sciences • 25-35 min

Category	Human Sciences	Time	25-35 min
Difficulty	Medium	ID	ultimatum-game

Big idea: Humans do not always maximise narrow utility.

Student-Facing Prompt

Inspect Ultimatum Game Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Ultimatum Game Stimulus A	Student-facing stimulus 1: Students split ten tokens and record accept/reject decisions. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Ultimatum Game Stimulus B	Student-facing stimulus 2: Offer cards showing fair, unfair, strategic, and emotional proposals. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Use this for comparison, a second round, or a partner challenge.
Ultimatum Game Reveal / Twist	Student-facing stimulus 3: A graph comparing rational-choice prediction with actual class choices. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.	Teacher reveal: connect the changed response to the big idea — Humans do not always maximise narrow utility.

Actual Lesson Questions

#	Question for students
1	After inspecting Ultimatum Game Stimulus A, what is your first knowledge claim?
2	Which exact detail from Ultimatum Game Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Can human behaviour be predicted scientifically?

Ready-to-Use Examples

- Students split ten tokens and record accept/reject decisions.
- Offer cards showing fair, unfair, strategic, and emotional proposals.
- A graph comparing rational-choice prediction with actual class choices.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Human Sciences.

Activity Steps

1. Pair students. Player A divides 10 tokens between A and B.
2. Player B accepts or rejects. If rejected, both get nothing.
3. Collect anonymous results.
4. Discuss why people reject unfair offers.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can human behaviour be predicted scientifically?
- Are humans rational knowers?
- How do fairness, emotion and culture affect decision-making?

Knowledge Questions

- Can human behaviour be predicted scientifically?
- Are humans rational knowers?
- How do fairness, emotion and culture affect decision-making?

Transfer Example

For example, run this activity with students split ten tokens and record accept/reject decisions. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...